

MIAE 221 · Rapid Review Sheet · Part 8

Phase Diagrams & Iron-Carbon Systems

1. Gibbs Phase Rule

- **Condensed Rule** ($P = 1 \text{ atm}$): $P + F = C + 1$
 - For binary system ($C = 2$): $P + F = 3 \implies F = 3 - P$
 - Single Phase ($P = 1 \implies F = 2$, Bivariant): Both T and composition can vary.
 - Two Phases ($P = 2 \implies F = 1$, Univariant): Selecting T fixes both phase compositions!
 - Three Phases ($P = 3 \implies F = 0$, Invariant): Invariant point (T and compositions fixed).
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2. Tie Line & Inverse Lever Rule

In any two-phase region at temperature T_0 for alloy composition C_0 :

1. **Compositions:** Read directly from the horizontal tie-line endpoints (C_α and C_L).
2. **Phase Fractions (Inverse Lever Rule):**

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{\text{Opposite Arm}}{\text{Total Length}}, \quad W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{\text{Opposite Arm}}{\text{Total Length}}$$

- Check: $W_\alpha + W_L = 1.00$ (100).
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3. Binary Eutectic Invariant Reactions

- **Eutectic Reaction:** $L \xrightarrow{\text{cool}} \alpha + \beta$ (Liquid solidifies into two distinct solids).
- **Pb-Sn System:** Eutectic at 183°C and 61.9 wt ($C_\alpha = 18.3$, $C_\beta = 97.8$).
- **Primary (Proeutectic) vs Eutectic Microconstituents** ($C_0 < 61.9$):

$$W_{\alpha'} = \frac{61.9 - C_0}{61.9 - 18.3}, \quad W_{\text{eutectic}} = 1 - W_{\alpha'} = \frac{C_0 - 18.3}{61.9 - 18.3}$$

- **Other Invariant Reactions:**
 - **Eutectoid:** $\text{Solid}_1 \rightarrow \text{Solid}_2 + \text{Solid}_3$ ($\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$)
 - **Peritectic:** $L + \text{Solid}_1 \rightarrow \text{Solid}_2$ ($L + \delta \rightarrow \gamma$)
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4. Iron-Carbon (Fe-Fe₃C) Phase Diagram

- **Allotropes of Iron:**
 - α -Ferrite: BCC, max C solubility 0.022 wt at 727°C . Soft, ductile.
 - γ -Austenite: FCC, max C solubility 2.14 wt at 1147°C . Non-magnetic.

○

Why FCC dissolves $100\times$ more C

: Octahedral voids in FCC are larger and symmetric ($r \approx 0.053$ nm vs 0.019 nm in BCC).

○ Cementite (Fe_3C): Intermetallic, 6.70 wt, hard and brittle.

• **Eutectoid Reaction** (727°C , 0.76 wt):

$$\gamma(0.76$$

○ Pearlite is a two-phase lamellar microconstituent (~ 88).

• **Hypoeutectoid Steels** ($C_0 < 0.76$):

- Proeutectoid ferrite: $W_{\alpha'} = \frac{0.76 - C_0}{0.76 - 0.022}$
- Pearlite: $W_p = 1 - W_{\alpha'} = \frac{C_0 - 0.022}{0.76 - 0.022}$
- Total ferrite: $W_{\alpha, \text{total}} = \frac{6.70 - C_0}{6.70 - 0.022}$